TESTBENCH:

```
module tb_toplayici;
   reg [3:0] in_a;
   reg [3:0] in_b;
   reg c_in;
   wire [3:0] sum_out;
   wire c_out;
   toplayici_4bit uut (
      .in_a(in_a),
      .in_b(in_b),
      .c_in(c_in),
      .sum_out(sum_out),
      .c_out(c_out)
   );
   initial begin
      in_a = 4'd0; in_b = 4'd0; c_in = 1'b0;
      #15;
      in_a = 4'b0101; in_b = 4'b0100; c_in = 1'b0;
      #15;
      in_a = 4'b1100; in_b = 4'b0111; c_in = 1'b0;
      #15;
      in_a = 4'b1000; in_b = 4'b0110; c_in = 1'b1;
      #15;
      $finish;
   end
   initial begin
      $monitor("T=%0t ns | A=%d (0b%b), B=%d (0b%b), Cin=%b | Sonuc=%d (0b%b), 
Cout=%b",
```

```
            $time, in_a, in_a, in_b, in_b, c_in, sum_out, sum_out, c_out);
    end
endmodule
```

DESİGN:

```
module toplayici_4bit (
    input [3:0] in_a,
    input [3:0] in_b,
    input c_in,
    output [3:0] sum_out,
    output c_out
);
    assign {c_out, sum_out} = in_a + in_b + c_in;
endmodule
```

ÇIKTI:

```
T=0 ns | A= 0 (0b0000), B= 0 (0b0000), Cin=0 | Sonuc= 0 (0b0000), Cout=0
T=15 ns | A= 5 (0b0101), B= 4 (0b0100), Cin=0 | Sonuc= 9 (0b1001), Cout=0
T=30 ns | A=12 (0b1100), B= 7 (0b0111), Cin=0 | Sonuc= 3 (0b0011), Cout=1
T=45 ns | A= 8 (0b1000), B= 6 (0b0110), Cin=1 | Sonuc=15 (0b1111), Cout=0
```